SDMMC SDIO eMMC Developer Guide

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Preface

Overview

Product Version

Chipset	Kernel Version
Full Series	4.4, 4.19, 5.10, 6.1

Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Revision History

Version	Author	Date	Change Description
V1.0.0	Lin Tao	2017-12-15	Initial version
V1.1.0	Lin Tao	2019-11-12	Revised for 4.19 kernel
V1.1.1	Huang Ying	2021-05-25	Modified format, added copyright information
V1.2.0	Zhao Yifeng	2022-09-26	Added SD card and JTAG multiplexing issue
V1.3.0	Lin Tao	2023-06-05	Added explanation of SD card leakage issue
V1.4.0	Lin Tao	2024-11-05	Supplemented mode switching test and mmc_test compatibility test items; corrected some attributes and descriptions

Contents

1. DTS Configuration

1.1 SDMMC DTS Configuration Guide

1. `max-frequency = <150000000>;`

This configuration sets the operating frequency of the SD card. Although set to 150MHz, adjustments may be necessary depending on the different modes of the SD card. Users do not need to be concerned with this part, as the actual operating frequency and module relationship will be handled by the associated software. The maximum does not exceed 150MHz.

2. `supports-sd;`

This configuration identifies the slot as an SD card function, which is a required addition. Without it, the SD card cannot be initialized. Note that after kernel 4.19, this configuration is no longer used, and instead, a combination of `no-sdio` and `no-mmc` is used to indicate that this card slot does not support SDIO and MMC, and is dedicated to SD functionality.

3. `bus-width = <4>;`

This configuration identifies the bus width required for the SD card. The SD card supports up to a 4-line mode, and if not configured, it defaults to a 1-line mode. Additionally, only the values 1 and 4 are supported; configuring any other value will be considered illegal and will force the use of a 1-line mode.

4. `cap-mmc-highspeed; cap-sd-highspeed;`

These configurations indicate that the slot supports highspeed SD cards. If not configured, it implies that highspeed SD cards are not supported.

5. Configuring SD3.0 Usage

First, ensure that the chip supports SD3.0 mode and configure the IO power supply of the vqmmc route for the SDMMC controller, and add the following SD3.0 speed modes:

```
sd-uhs-sdr12: Clock frequency does not exceed 24M
sd-uhs-sdr25: Clock frequency does not exceed 50M
sd-uhs-sdr50: Clock frequency does not exceed 100M
sd-uhs-ddr50: Clock frequency does not exceed 50M, and uses dual-edge sampling
sd-uhs-sdr104: Clock frequency does not exceed 208M
```

6. Configuring the 3V3 Power Supply for SD Card Devices

If the hardware uses the power control pin of the chip's SDMMC controller as the default power control pin for the SD card: sdmmc_pwren, then only need to configure the pinctrl as the functional pin for sdmmc_pwren and reference it in the default pinctrl of the sdmmc node, for example, using RK312X as an example:

```
sdmmc_pwren: sdmmc-pwren {
    rockchip,pins = <1 RK_PB6 1 &pcfg_pull_default>;
};

pinctrl-0 = <&sdmmc_pwr &sdmmc_clk &sdmmc_cmd &sdmmc_bus4>;
```

If the hardware uses another GPIO as the 3V3 power control pin for the SD card device, it needs to be defined as a regulator for use and referenced in the vmmc-supply within the sdmmc node, for example:

```
sdmmc_pwr: sdmmc-pwr {
    rockchip,pins = <7 11 RK_FUNC_GPIO &pcfg_pull_none>;
};

vcc_sd: sdmmc-regulator {
    compatible = "regulator-fixed";
    gpio = <&gpio7 11 GPIO_ACTIVE_LOW>;
    pinctrl-names = "default";
    pinctrl-0 = <&sdmmc_pwr>;
    regulator-name = "vcc_sd";
    regulator-min-microvolt = <3300000>;
    regulator-max-microvolt = <3300000>;
    startup-delay-us = <100000>;
    vin-supply = <&vcc_io>;
};

&sdmmc {
    vmmc-supply = <&vcc_sd>;
};
```

7. Configuring SD Card Hot-Swap Detection Pins

If the detection pin is directly connected to the chip's SDMMC controller's sdmmc_cd pin, then simply configure the pin as a functional pin and reference it in the default pinctrl of the sdmmc node.

If the detection pin uses another GPIO, it needs to be configured using cd-gpios within the sdmmc node, for example:

`cd-gpios = <&gpio4 24 GPIO_ACTIVE_LOW>;`

If a GPIO detection pin is used, but an inverted detection method is required (i.e., the detection pin is high when the SD card is inserted), then add:

`cd-inverted;`

1.2 SDIO DTS Configuration Guide

1. `max-frequency = <150000000>;`

This item is the same as the SD card configuration, with the maximum operating frequency not exceeding 150MHz; if an SDIO2.0 device is connected, it will automatically be compatible with a maximum of 50MHz, and if an SDIO3.0 device is connected, it will support up to 150MHz.

2. `supports-SDIO;`

This configuration identifies this slot as an SDIO function, which is a required addition. Otherwise, the SDIO peripheral cannot be initialized. Note that after the 4.19 kernel, this configuration is no longer used, and the combination of `no-sd` and `no-mmc` is used to indicate that this card slot does not support SDMMC and MMC, and is dedicated to SDIO functionality.

3. `bus-width = <4>;`

This configuration is the same as the SD card functionality.

4. `cap-sd-highspeed;`

This configuration is the same as the SD card functionality, and as an SDIO peripheral, it also distinguishes whether it is a highspeed SDIO peripheral.

5. `cap-sdio-irq;`

This configuration indicates whether the SDIO peripheral (usually WiFi) supports SDIO interrupts. If your peripheral is an OOB interrupt, please do not include this item. For the type of interrupt supported, please contact the WiFi manufacturer to confirm.

6. `keep-power-in-suspend;`

This configuration indicates whether sleep power-off is supported, and this option should be added by default. WiFi generally has deep wake-up requirements.

7. `mmc-pwrseq = <&sdio_pwrseq>;`

This item is the power control for the SDIO peripheral (usually WiFi) and is a required item, otherwise, WiFi cannot power up and work. Please refer to the example below, and the selection of crystal clock and reset-enable GPIO should be configured according to the actual board-level hardware requirements.

```
        sdio_pwrseq:sdio-pwrseq {
                compatible ="mmc-pwrseq-simple";
                clocks = <&rk808 1>;
                clock-names ="ext_clock";
                pinctrl-names ="default";
                pinctrl-0 =<&wifi_enable_h>;
                /*
                * On the module itself this is one of these (depending
                * on the actual card populated):
                * - SDIO_RESET_L_WL_REG_ON
                * - PDN (power down when low)
                */
                reset-gpios = <&gpio0 10GPIO_ACTIVE_LOW>; /* GPIO0_B2 */
        };
```

8. `non-removable;`

This item indicates that the slot is a non-removable device and is a required addition for SDIO devices.

9. `num-slots = <4>;`

This item is the same as the SD card configuration.

10. `sd-uhs-sdr104;`

This configuration determines whether the SDIO device supports the SDIO3.0 mode. The prerequisite is that the IO voltage of the WiFi is 1.8v.

1.3 eMMC's DTS Configuration

1. `max-frequency = <150000000>;`

This setting is the same as the SD card configuration, with the maximum operating frequency not exceeding 150MHz; if operating in eMMC Highspeed mode, it automatically downshifts to 50MHz compatibility, and if operating in eMMC HS200 mode, it supports up to 150MHz.

2. `supports-emmc;`

This configuration identifies this slot as an eMMC function, which is a mandatory addition. Without it, the eMMC peripheral cannot be initialized. Note that after the 4.19 kernel, this configuration is no longer used, and instead, the combination of `no-sd` and `no-sdio` is used to indicate that this card slot does not support SDMMC and SDIO, dedicated to eMMC functionality.

3. `bus-width = <4>;`

This configuration is the same as the SD card functionality.

4. `mmc-ddr-1_8v;`

This configuration indicates support for the 50MDDR mode.

5. `mmc-hs200-1_8v;`

This configuration indicates support for the HS200 mode.

6. `mmc-hs400-1_8v; mmc-hs400-enhanced-strobe`

These two configurations indicate support for the HS400 mode and the HS400ES mode. Please refer to the chip specifications for specific chip support details.

7. `non-removable;`

This item indicates that the slot is for a non-removable device. This is a mandatory addition.

2. Frequency and Mode Switching

1. Mode switching can be achieved using the relevant items in the aforementioned DTS configuration.
2. If you wish to switch modes without modifying the firmware, a dynamic switching method can be employed.

```
The kernel version 6.1 comes with the following commits by default; other platforms
need to apply them.
commit 0c11160a3c03bcf41fd1217d5e44e68ff078c54b
Author: Shawn Lin <shawn.lin@rock-chips.com>
```

```
Date:   Tue Nov 5 18:05:50 2024 +0800

    mmc: debugfs: Allow more host caps to be modified
    Change-Id: I2d5df7bc95eaa4a380b5107364aff8df3a759788

commit 478cdcd25cba8be5cfcf56b503b0e070f5aef245
Author: Vincent Whitchurch <vincent.whitchurch@axis.com>
Date:   Fri Sep 29 09:45:09 2023 +0200

    UPSTREAM: mmc: debugfs: Allow host caps to be modified
    Change-Id: I6ea2fc093234785e05d05128556c60ef42da02a4

commit a5a3ea5e45d09cff84c04b515b5b71d4dd76524e
Author: Vincent Whitchurch <vincent.whitchurch@axis.com>
Date:   Fri Sep 29 09:45:08 2023 +0200

    UPSTREAM: mmc: core: Always reselect card type
    Change-Id: Ie80514ade3b267901818f477d74d91cf7c13e103
```

2.1 SD/SDIO Dynamic Switching

1. **DTS configures according to the maximum supported rate mode**. Identify the MMC device ID and address to be modified from the log, taking a specific TF card as an example.

```
[    5.739929] mmc_host mmc1: Bus speed (slot 0) = 400000Hz (slot req 400000Hz, actual
400000HZ div = 0)
[    5.941037] mmc_host mmc1: Bus speed (slot 0) = 198000000Hz (slot req 200000000Hz,
actual 198000000HZ div = 0)
[    5.941682] dwmmc_rockchip 2a310000.mmc: Successfully tuned phase to 90
[    5.941705] mmc1: new ultra high speed SDR104 SDHC card at address aaaa
[    5.943229] mmcblk1: mmc1:aaaa SC32G 29.7 GiB
```

We can obtain three pieces of information:

- The MMC device ID is 1, i.e., **mmc1**
- The address of the TF card is **aaaa**, and the path combination is **mmc1:aaaa**
- The current operating mode is **sdr104** mode, with a frequency of **198MHz**

2. Since this device is mmc1, execute `cd /sys/kernel/debug/mmc1` to enter the corresponding directory. If it were mmc0, you would enter the mmc0 directory, and so on.
3. If you wish to switch the frequency to 100MHz in the current mode, use the `echo 100000000 > clock` command, and so on.

```
root@rk3576-buildroot:/sys/kernel/debug/mmc1# echo 100000000 > clock
[   86.451685] mmc_host mmc1: Bus speed (slot 0) = 100000000Hz (slot req 100000000Hz,
actual 100000000HZ div = 0)
```

4. Switch to High speed mode, with a default frequency of 50MHz.

```
#Execute the following commands in the sys/kernel/debug/mmc1 directory:
echo $(($(cat caps) | (1 << 7))) > caps
echo on > /sys/bus/mmc/devices/mmc1\:aaaa/power/control
echo auto > /sys/bus/mmc/devices/mmc1\:aaaa/power/control
echo $(($(cat caps) & ~(1 << 19))) > caps
echo on > /sys/bus/mmc/devices/mmc1\:aaaa/power/control

#You can see the switch has occurred
[   78.304578] mmc_host mmc1: Bus speed (slot 0) = 400000Hz (slot req 400000Hz, actual
400000HZ div = 0)
[   78.501126] mmc_host mmc1: Bus speed (slot 0) = 50000000Hz (slot req 50000000Hz,
actual 50000000HZ div = 0)

#Execute the following command to confirm the switch was successful
grep timing ios
timing spec:       2 (sd high-speed) #High speed mode now
```

5. Switch to Default speed mode, with a default frequency of 25MHz.

```
#Execute the following commands in the sys/kernel/debug/mmc1 directory:
echo $(($(cat caps) | (1 << 7))) > caps
echo on > /sys/bus/mmc/devices/mmc1\:aaaa/power/control
echo auto > /sys/bus/mmc/devices/mmc1\:aaaa/power/control
echo $(($(cat caps) & ~(1 << 2))) > caps
echo on > /sys/bus/mmc/devices/mmc1\:aaaa/power/control

#You can see the switch has occurred
[  287.590885] mmc_host mmc1: Bus speed (slot 0) = 400000Hz (slot req 400000Hz, actual
400000HZ div = 0)
[  287.784531] mmc_host mmc1: Bus speed (slot 0) = 25000000Hz (slot req 25000000Hz,
actual 25000000HZ div = 0)

#Execute the following command to confirm the switch was successful
grep timing ios
timing spec:    0 (legacy) #Default speed mode now
```

2.2 Dynamic Switching of eMMC

1. **DTS is configured according to the maximum supported rate mode**. Identify the MMC device ID and address to be modified from the log, taking RK3588 eMMC as an example.

```
[    2.260445] mmc0: Command Queue Engine enabled
[    2.260458] mmc0: new HS400 Enhanced strobe MMC card at address 0001
[    2.260740] mmcblk0: mmc0:0001 BJTD4R 29.1 GiB
```

We can obtain three pieces of information:

- The MMC device ID is 0, i.e., **mmc0**
- The eMMC device address is **0001**, and the path combination is **mmc0:0001**

- The current operating mode is **HS400** mode, with the platform eMMC default frequency **198MHz**

2. Since this device is mmc0, execute `cd /sys/kernel/debug/mmc0` to enter the corresponding directory. If it were mmc1, you would enter the mmc1 directory, and so on.
3. If you wish to switch frequencies within the current mode, use the `echo 100000000 > clock` command, and so on.

```
# First confirm the default frequency and mode
root@rk3588-buildroot:/sys/kernel/debug/mmc0# grep clock ios
clock:          200000000 Hz
actual clock:   198000000 Hz  #Default 198MHz
root@rk3588-buildroot:/sys/kernel/debug/mmc0# grep timing ios
timing spec:    10 (mmc HS400 enhanced strobe) #HS400es mode

# Begin frequency switching
root@rk3588-buildroot:/sys/kernel/debug/mmc0# echo 100000000 > clock
root@rk3588-buildroot:/sys/kernel/debug/mmc0# grep clock ios
clock:          100000000 Hz
actual clock:   100000000 Hz  #Switched to 100MHz
root@rk3588-buildroot:/sys/kernel/debug/mmc0# grep timing ios
timing spec:    10 (mmc HS400 enhanced strobe) #Still HS400es mode
```

4. Switch to HS200 mode, with a default frequency of 198MHz.

```
# Execute the following commands in the sys/kernel/debug/mmc0 directory:
echo $(($(cat caps) | (1 << 7))) > caps
echo on > /sys/bus/mmc/devices/mmc0:0001/power/control
echo auto > /sys/bus/mmc/devices/mmc0:0001/power/control
echo $(($(cat caps2) & ~(1 << 15))) > caps2
echo on > /sys/bus/mmc/devices/mmc0:0001/power/control

# Execute the following command to confirm the switch was successful
grep timing ios
timing spec:       9 (mmc HS200) #HS200 mode now
```

5. Switch to High speed mode, with a default frequency of 50MHz.

```
# Execute the following commands in the sys/kernel/debug/mmc0 directory:
echo $(($(cat caps) | (1 << 7))) > caps
echo on > /sys/bus/mmc/devices/mmc0:0001/power/control
echo auto > /sys/bus/mmc/devices/mmc0:0001/power/control
echo $(($(cat caps) & ~(1 << 1))) > caps
echo on > /sys/bus/mmc/devices/mmc0:0001/power/control

# Execute the following command to confirm the switch was successful
grep timing ios
timing spec:    1 (mmc high-speed) #High speed mode now
```

2.3 Additional Information

In addition to frequency and mode switching, line width switching (downward switching of multiple line widths may cause automatic downgrade of modes), CMD23 support switch, CQE and DCMD switches are also supported. Taking the mmc0:0001 device as an example:

```
# Execute the following commands in sequence in the sys/kernel/debug/mmc0 directory:
echo $(($(cat caps) | (1 << 7))) > caps
echo on > /sys/bus/mmc/devices/mmc0\:0001/power/control
echo auto > /sys/bus/mmc/devices/mmc0\:0001/power/control

# Select the following switching configurations as needed
echo $(($(cat caps) & ~(1 << 6))) > caps  # Remove 8-bit support
echo $(($(cat caps) | (1 << 6))) > caps  # Add 8-bit support
echo $(($(cat caps) & ~(1 << 0))) > caps  # Remove 4-bit support
echo $(($(cat caps) | (1 << 0))) > caps  # Add 4-bit support
echo $(($(cat caps) & ~(1 << 30))) > caps  # Remove cmd23 support
echo $(($(cat caps) | (1 << 30))) > caps  # Add cmd23 support
echo $(($(cat caps) & ~(1 << 30))) > caps  # Remove cmd23 support
echo $(($(cat caps) | (1 << 30))) > caps  # Add cmd23 support
echo $(($(cat caps2) & ~(1 << 23))) > caps2  # Remove cqe support
echo $(($(cat caps2) | (1 << 23))) > caps2  # Add cqe support
echo $(($(cat caps2) & ~(1 << 24))) > caps2  # Remove cqe dcmd support
echo $(($(cat caps2) | (1 << 24))) > caps2  # Add cqe dcmd support

# Execute this again
echo on > /sys/bus/mmc/devices/mmc0\:0001/power/control

# Confirm ios information, check the output results
cat ios
```

The bit information supported by caps and caps2 can be queried in the kernel include/linux/mmc/host.h file. The currently supported caps and caps2 are:

caps	caps2
MMC_CAP_AGGRESSIVE_PM	MMC_CAP2_HS400_1_8V
MMC_CAP_SD_HIGHSPEED	MMC_CAP2_HS400_1_2V
MMC_CAP_MMC_HIGHSPEED	MMC_CAP2_HS400_ES
MMC_CAP_UHS	MMC_CAP2_HS200_1_8V_SDR
MMC_CAP_DDR	MMC_CAP2_HS200_1_2V_SDR
MMC_CAP_4_BIT_DATA	MMC_CAP2_CQE
MMC_CAP_8_BIT_DATA	MMC_CAP2_CQE_DCMD
MMC_CAP_CMD23	

If there are additional configurations required for the project, you can follow the aforementioned `commit 0c11160a3c03bcf41fd1217d5e44e68ff078c54b` submission to add them yourself.

3. MMC Test Compatibility Testing

It is recommended that products dependent on SD cards be integrated into the SD card-specific compatibility testing.

3.1 Kernel Configuration Confirmation

```
Please ensure the following configuration CONFIG_MMC_TEST=y is added:
Device Drivers  -->
   <*> MMC/SD/SDIO card support  -->
   [*]   MMC host test driver

If selected as a module, please insmod mmc_test.ko after booting
```

3.2 Usage

```
1. List all mmc devices
ls /sys/bus/mmc/devices/

2. Identify the mmc device to be tested, for example, an SD card, and query its path
combination from the log as
mmc1:aaaa

3. Remove the SD card from the system
echo 'mmc1:aaaa' > /sys/bus/mmc/drivers/mmcblk/unbind

4. Bind the SD card to mmc_test
echo 'mmc1:aaaa' > /sys/bus/mmc/drivers/mmc_test/bind

5. At this point, you can see the following log
mmc_test mmc1:aaaa: Card claimed for testing.

6. Check the available test items
cat /sys/kernel/debug/mmc1/mmc1:aaaa/testlist

7. Select the test item according to the test requirements, for the meaning of N, please
refer to the test description below
echo N > /sys/kernel/debug/mmc1/mmc1:aaaa/test
```

3.3 Output Results

```
mmc1: Result: OK                    ---Test Completed
mmc1: Result: FAILED                ---Test Failed
mmc1: Result: UNSUPPORTED (by host) ---Controller Configuration Unsupported
mmc1: Result: UNSUPPORTED (by card) ---Card Unsupported
```

3.4 Test Description

N Value	Test Name	Description
0	Run all tests	Execute all tests
1	Basic write (no data verification)	Write test without data verification
2	Basic read (no data verification)	Read test without data verification
3	Basic write (with data verification)	Write test with data verification
4	Basic read (with data verification)	Read test with data verification
5	Multi-block write	Multi-block writing
6	Multi-block read	Multi-block reading
7	Power of two block writes	Even power block writes
8	Power of two block reads	Even power block reads
9	Weird sized block writes	Various odd block sizes writing
10	Weird sized block reads	Various odd block sizes reading
11	Badly aligned write	Misaligned address single block write
12	Badly aligned read	Misaligned address single block read
13	Badly aligned multi-block write	Misaligned address multi-block write
14	Badly aligned multi-block read	Misaligned address multi-block read
15	Proper xfer_size at write (start failure)	Intentionally create an incorrect transfer type, e.g., using single block transfer command for multi-block write
16	Proper xfer_size at read (start failure)	Intentionally create an incorrect transfer type, e.g., using single block transfer command for multi-block read
17	Proper xfer_size at write (midway failure)	Same as 15, but insert write error in the middle of multi-block transfer
18	Proper xfer_size at read (midway failure)	Same as 16, but insert read error in the middle of multi-block transfer
19	Highmem write	Execute single block write data using High memory page
20	Highmem read	Execute single block read data using High memory page

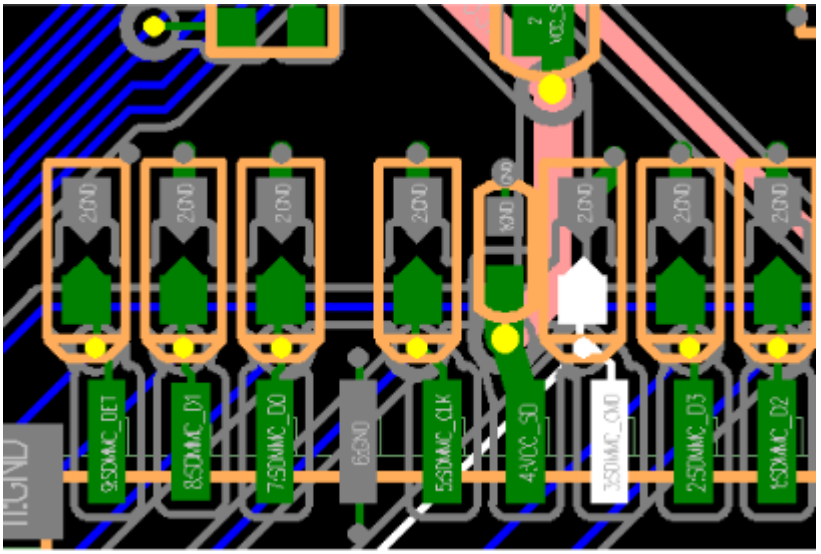
N Value	Test Name	Description
21	Multi-block highmem write	Execute multi-block write data using High memory page
22	Multi-block highmem read	Execute multi-block read data using High memory page
23	Best-case read performance	Perform 512K sequential read, continuous address, without sg mode
24	Best-case write performance	Perform 512K sequential write, continuous address, without sg mode
25	Best-case read performance into scattered pages	Perform 512K sequential read, enable sg mode
26	Best-case write performance from scattered pages	Perform 512K sequential write, enable sg mode
27	Single read performance by transfer size	Single block read performance
28	Single write performance by transfer size	Single block write performance
29	Single trim performance by transfer size	Single block erase performance
30	Consecutive read performance by transfer size	Continuous read performance
31	Consecutive write performance by transfer size	Continuous write performance
32	Consecutive trim performance by transfer size	Continuous erase performance
33	Random read performance by transfer size	Random read performance
34	Random write performance by transfer size	Random write performance
35	Large sequential read into scattered pages	Large continuous read test using sg mode
36	Large sequential write from scattered pages	Large continuous write test using sg mode
37	Write performance with blocking req 4k to 4MB	Test blocking write performance with different transfer lengths from 4K to 4MB

N Value	Test Name	Description
38	Write performance with non-blocking req 4k to 4MB	Test non-blocking write performance with different transfer lengths from 4K to 4MB
39	Read performance with blocking req 4k to 4MB	Test blocking read performance with different transfer lengths from 4K to 4MB
40	Read performance with non-blocking req 4k to 4MB	Test non-blocking read performance with different transfer lengths from 4K to 4MB
41	Write performance blocking req 1 to 512 sg elems	Test blocking write performance with 1 to 512 sg elements
42	Write performance non-blocking req 1 to 512 sg elems	Test non-blocking write performance with 1 to 512 sg elements
43	Read performance blocking req 1 to 512 sg elems	Test blocking read performance with 1 to 512 sg elements
44	Read performance non-blocking req 1 to 512 sg elems	Test non-blocking read performance with 1 to 512 sg elements
45	Reset test	Reset test item
46	Commands during read - no Set Block Count (CMD23)	Send various commands during read (non-CMD23 method)
47	Commands during write - no Set Block Count (CMD23)	Send various commands during write (non-CMD23 method)
48	Commands during read - use Set Block Count (CMD23)	Send various commands during read (CMD23 method)
49	Commands during write - use Set Block Count (CMD23)	Send various commands during write (CMD23 method)
50	Commands during non-blocking read - use Set Block Count (CMD23)	Send various commands during non-blocking read (CMD23 method)
51	Commands during non-blocking write - use Set Block Count (CMD23)	Send various commands during non-blocking write (CMD23 method)

4. Troubleshooting Common Issues

4.1 Hardware Issue Analysis

1. SD Card



From left to right, they are:

DET ---- Detection pin

DATA1 ---- Data line

DATA0

GND

CLK ---- Clock

VCC_SD ---- SD card power supply

VCCIO_SD ---- Data line IO power supply

CMD ---- Command line

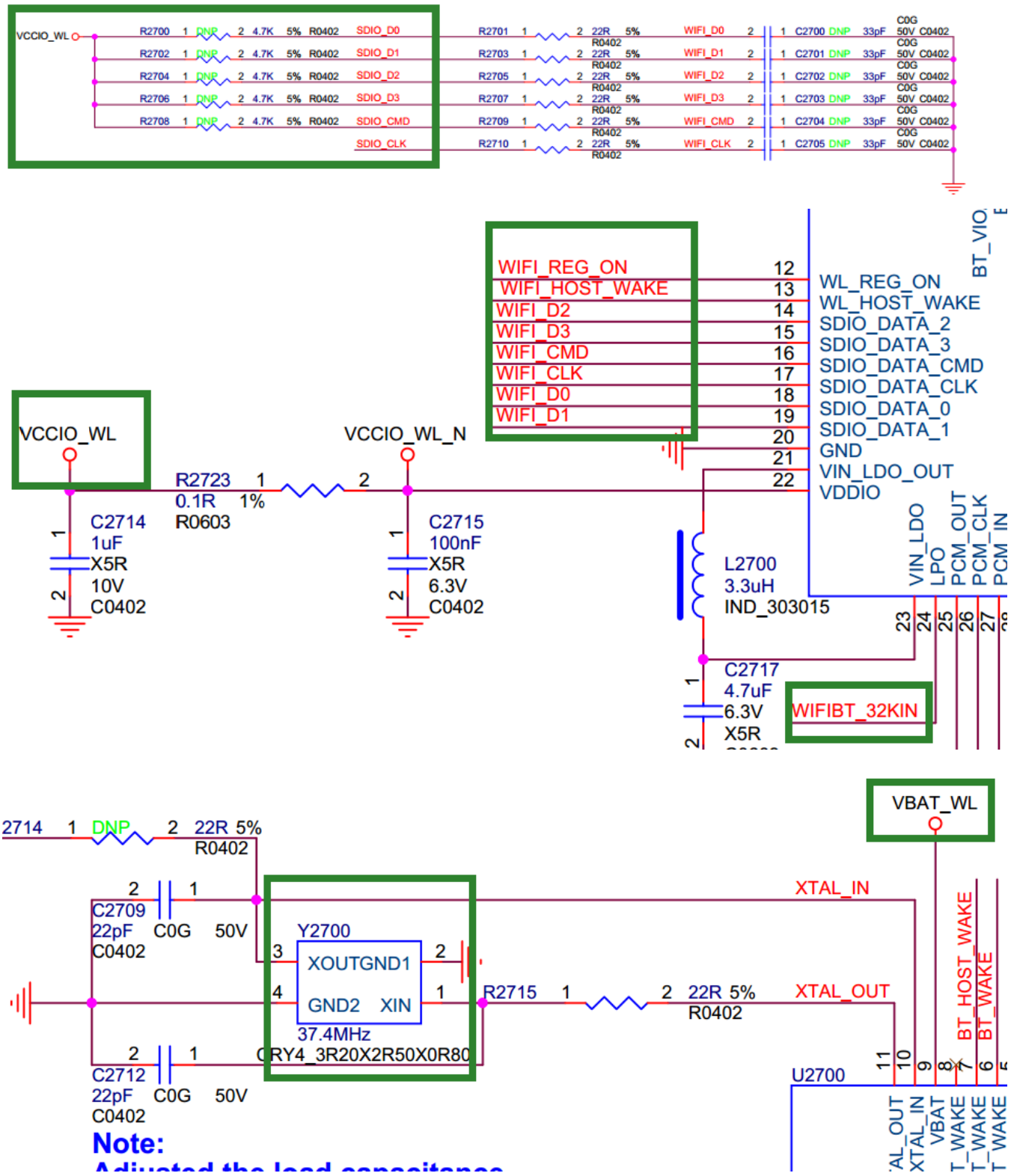
DATA3

DATA2

Except for DET/CLK/GND, the other DATA0-3/VCC_SD/VCCIO_SD/CMD must all be around 3.3v, and the minimum should not be lower than 3v; DET pin is low when inserted and high when removed; The voltage of DATA0-3/CMD is supplied by VCCIO_SD, so DATA0-3/CMD must be consistent with VCCIO_SD, and VCC_SD and VCCIO_SD must be consistent (NOTE: SD 3.0 requires VCCIO_SD to be 1.8v);

If the power supply of VCC_SD/VCCIO_SD is a long-term supply, then please ensure that VCC_SD and VCCIO_SD will not collapse when the card is inserted or removed;

2. SDIO



First, let's look at the hardware: the main parts are all within the green square

WIFI_D0~3: Data lines, normally high, voltage depends on the voltage of VCCIO_WL;

WIFI_CMD: Command line, normally high, voltage depends on the voltage of VCCIO_WL;

WIFI_CLK: Clock, normally low, voltage depends on the voltage of VCCIO_WL;

VBAT_WL: WIFI module power supply, always high, power supply needs to print 3.3v;

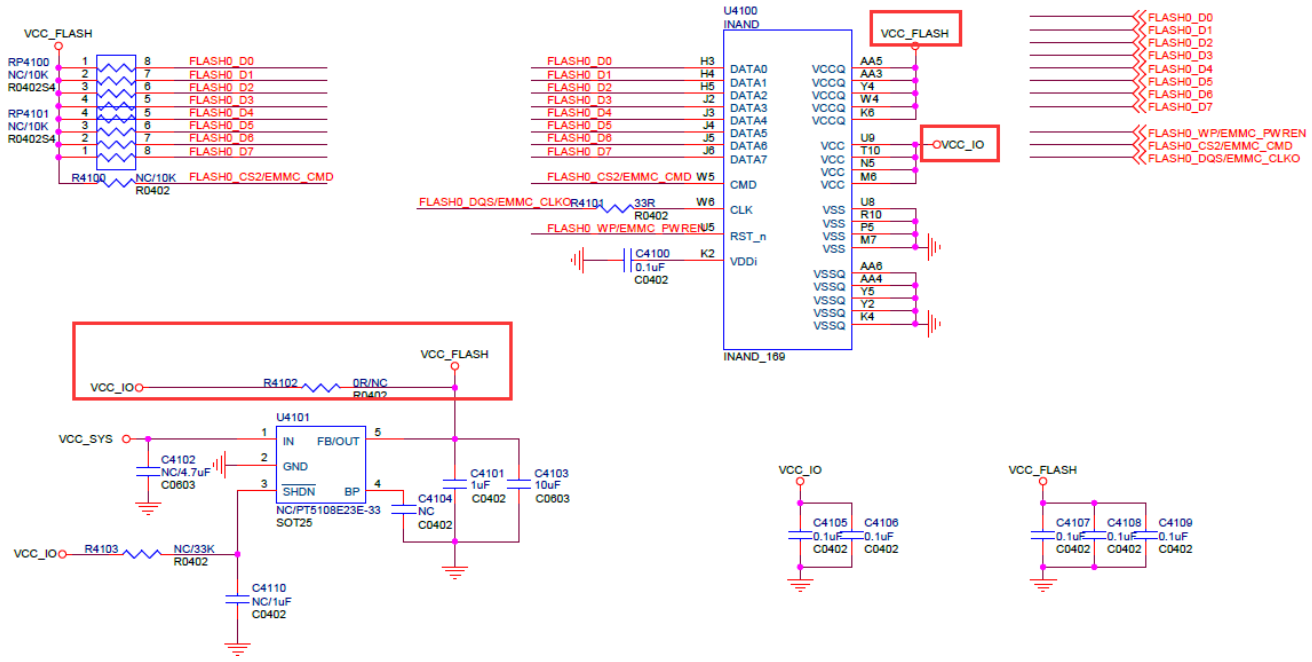
VCCIO_WL: IO power supply for DATA/CMD/CLK, can be 3.3 or 1.8v, but SDIO3.0

must be 1.8v;

WIFI_REG_ON: 3.3v when working normally, 0v when WiFi is turned off;

Two crystal oscillators: 32K and 26M/37.4M, both will have waveform output when working normally;

3. eMMC



eMMC valid voltage combinations:

Table 199 — *e*•MMC voltage combinations

		V_{CCQ}		
		1.1 V-1.3 V	**1.70 V–1.95 V**	**2.7 V–3.6 V**
V_{CC}	**2.7 V–3.6 V**	Valid	Valid	Valid (1)
	1.7 V–1.95 V	Valid	Valid	NOT VALID

NOTE 1 V_{CCQ} (I/O) 3.3 V range is not supported in either HS200 or HS400 devices

VCC_FLASH corresponds to VCC;

VCC_IO corresponds to VCCQ;

Ensure that the voltages of eMMC_CMD/DATA0~7/VCC_IO are consistent (1.8 or 3.3v);

Ensure that the voltages of VCC_FLASH/VCC_IO remain stable during power-on, operation, or sleep and wake-up, without any collapse or excessive ripple;

If possible, measure the waveform quality of clk, cmd, and data to ensure the waveform is normal.

4.2 Waveform Analysis

The following diagram is the waveform timing chart when the SD card is recognized in mode (the same for sdio and emmc).

A brief explanation of how to identify an SD card: The controller sends out 48clk and carries 48bit data to the SD card, and the SD card needs to respond to the controller with 48clk and 48bit data; as shown in the figure below:

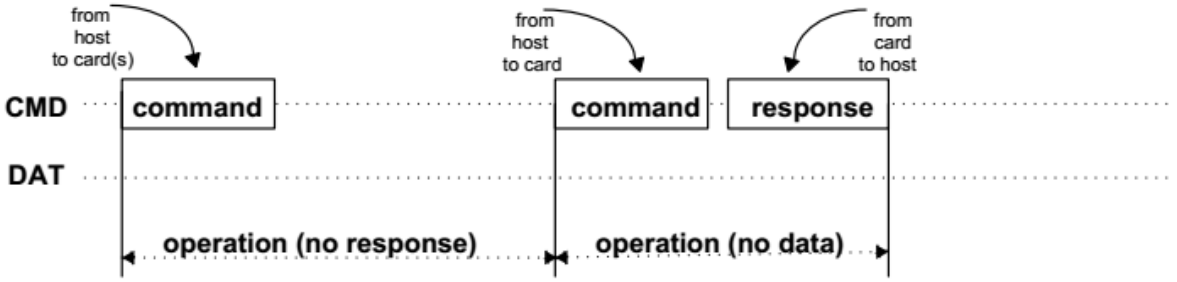
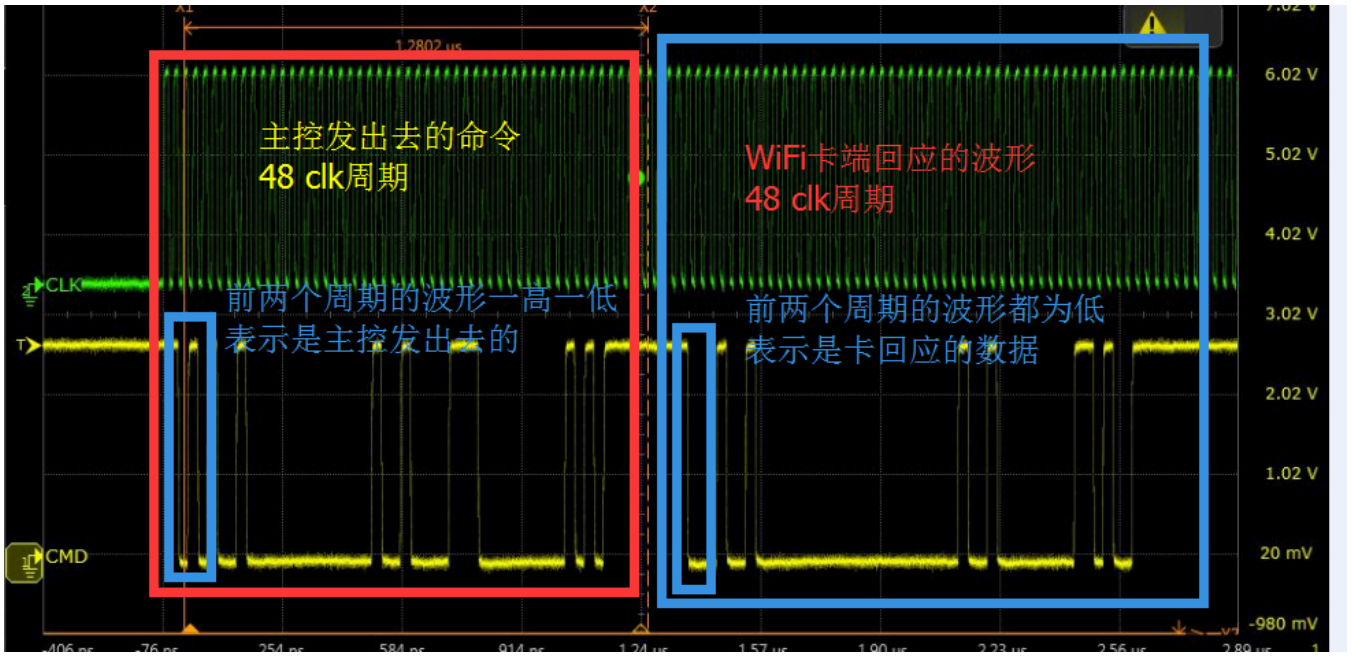


Figure 3-4: "no response" and "no data" Operations



Green: SDMMC_CLK

Yellow: SDMMC_CMD: SDMMC_CMD remains at a high level when idle;

The waveform sent by the controller: When the first two levels have one high and one low, it indicates the command sent by the controller;

The waveform response from the SD card: When the first two levels have two consecutive low levels, it indicates that the card side has responded;

Secondly, the controller and response generally include 48 bits of data, so 48 clks constitute a complete package. What needs to be confirmed is: after the controller sends out the command package, whether the SD card side responds.

4.3 LOG Analysis

1. Correctly Identify SD Card LOG

```
[  293.194013] mmc1: new high speed SDXC card at address 59b4
[  293.198185] mmcblk1: mmc1:59b4 00000 59.6 GiB
[  293.204351]  mmcblk1: p1
```

If such printouts are seen in the kernel, it indicates that the SD card has been correctly identified and has an available partition p1.

If the SD card device is not visible in the user interface or is unusable, please troubleshoot the user-space disk daemon, such as vold.

Additionally, manually verify if the partition is usable

```
mount -t vfat /dev/block/mmcblk1p1 /mnt
```

or

```
mount -t vfat /dev/block/mmcblk1 /mnt
```

Then go to the mnt directory to see if there are files from the SD card.

2. No card reading at boot, OK when hot-swapped during runtime: Most likely a power supply issue

For example: Disconnect all power sources and find that there is leakage to VCC_SD card when HDMI is plugged in; or test with an external power supply.

3. Mount Failure:

If you have seen the LOG in (1), but see the following mount failure LOG

```
[ 2229.405694] FAT-fs (mmcblk1p1): bogus number of reserved sectors
[ 2229.405751] FAT-fs (mmcblk1p1): Can't find a valid FAT filesystem
```

Please format the SD card to the FAT32 file system;

Or NTFS: make menuconfig to select support for the NTFS file system;

4. Probabilistic Non-Recognition:

```
mmc1: new high speed SD card at address b368
mmcblk1: mmc1:b368 SMI   486 MiB
[mmc1] Data transmission error !!!!  MINTSTS: [0x00002000]
dwmmc_rockchip ff0c0000.rksdmmc: data FIFO error (status=00002000)
mmcblk1: error -110 sending status command, retrying
need_retune:0,brq->retune_retry_done:0.
```

Reduce frequency and increase card detection delay to enhance power stability. If reducing frequency works OK, please check the hardware layout;

```
&sdmmc {
    card-detect-delay = <1200>;
}
```

5. TF card has been mounted, but cannot access the TF card directory, it seems to be a card file system issue, but the card can be accessed under Windows.

Please try to use fsck to repair the TF card.

6. Hardware issue, IO voltage abnormal

```
Workqueue: kmmcd mmc_rescan
[<c0013e24>] (unwind_backtrace+0x0/0xe0) from [<c001172c>] (show_stack+0x10/0x14)
[<c001172c>] (show_stack+0x10/0x14) from [<c04fa444>] (dw_mci_set_ios+0x9c/0x21c)
[<c04fa444>] (dw_mci_set_ios+0x9c/0x21c) from [<c04e7748>] (mmc_set_chip_select+0x18/0x1c)
[<c04e7748>] (mmc_set_chip_select+0x18/0x1c) from [<c04ebd5c>] (mmc_go_idle+0x94/0xc4)
[<c04ebd5c>] (mmc_go_idle+0x94/0xc4) from [<c0748d80>] (mmc_rescan_try_freq+0x54/0xd0)
[<c0748d80>] (mmc_rescan_try_freq+0x54/0xd0) from [<c04e85d0>] (mmc_rescan+0x2c4/0x390)
[<c04e85d0>] (mmc_rescan+0x2c4/0x390) from [<c004d738>] (process_one_work+0x29c/0x458)
[<c004d738>] (process_one_work+0x29c/0x458) from [<c004da88>] (worker_thread+0x194/0x2d4)
[<c004da88>] (worker_thread+0x194/0x2d4) from [<c0052fb4>] (kthread+0xa0/0xac)
[<c0052fb4>] (kthread+0xa0/0xac) from [<c000da98>] (ret_from_fork+0x14/0x3c)
1409..dw_mci_set_ios:  wait for unbusy timeout..... STATUS = 0x306 [mmc1]
```

Please check whether the voltage of the CMD line and DATA is at a high level under no-load conditions. And check whether the IO voltage is too low, and whether the IO voltage and power domain configuration are consistent. If it is an SDIO interface, it is recommended to troubleshoot VCCIO_WL voltage, VBAT_WL and WIFI_REG_ON as well as whether the crystal oscillator is normal. Also, try to troubleshoot whether the long wiring leads to poor waveform quality, and test by reducing frequency.

4.4 Other Issues

1. SD Card Works Normally in 1-Wire Mode but Reports Errors in 4-Wire Mode

Most SOCs' SD cards share lines with JTAG, and when no card is inserted, the SOC automatically switches the IO to JTAG functionality.

The EVB reference board is designed according to the SOC requirements, with SD DET low level indicating card insertion. Some customers may modify the schematics to design SD DET high level for card insertion, causing the SOC to misjudge and switch the IO to JTAG functionality.

Solution: Look for the corresponding chip GRF register definition and configure force_jtag as Disable to turn off the automatic switching of JTAG IO.

Reference code for RV1126:

```
diff --git a/arch/arm/mach-rockchip/rv1126/rv1126.c b/arch/arm/mach-rockchip/rv1126/rv1126.c
index 311310d3f2..29b694df9c 100644
--- a/arch/arm/mach-rockchip/rv1126/rv1126.c
+++ b/arch/arm/mach-rockchip/rv1126/rv1126.c
@@ -544,6 +544,9 @@ void board_debug_uart_init(void)
 #ifndef CONFIG_TPL_BUILD
 int arch_cpu_init(void)
 {
+       struct rv1126_grf * const grf = (void *)GRF_BASE;
```

```
+
+       writel(0x00100000, &grf->iofunc_con3);
        /*
         * CONFIG_DM_RAMDISK: for ramboot that without SPL.
         */
```

2. SD Card Leakage Causes Abnormal Card Operation

When the IO power supply of the SD module is uncontrollable, if a card is inserted before the 3V3 power supply is provided, leakage may occur from the IO, causing half-level voltage in the SD card's 3V3 power supply, which can lead to sporadic abnormalities in some cards, manifesting as poor platform compatibility. To solve the leakage issue, the driver will set the SD card IO to pull down when the card is inserted, release the leakage, and then power up the SD card's 3V3 power supply. After the SD card is powered up, the IO is restored to SD function pins and pull-up is set to avoid a series of abnormalities that may be caused by leakage. Customers with high compatibility requirements for SD cards, please confirm that the SDK kernel's drivers/mmc/host/dw_mmc.c file includes the submission "mmc: dw_mmc: Add normal and idle pinctrl control", and refer to the following example to modify your own DTS node, adding an idle mode for pinctrl, which should be configured according to the actual usage of the board with the required clk, cmd, and data lines:

```
--- a/arch/arm/boot/dts/rv1126-evb-v10.dtsi
+++ b/arch/arm/boot/dts/rv1126-evb-v10.dtsi
    wireless_wlan: wireless-wlan {
@@ -104,9 +99,14 @@ sdmmc_pwren: sdmmc-pwren {
            rockchip,pins = <0 RK_PA1 RK_FUNC_GPIO &pcfg_pull_none>;
        };

+       sdmmc0_idle_pins: sdmmc-idle-pins {
+           rockchip,pins =
+               <3 RK_PA2 RK_FUNC_GPIO &pcfg_pull_down>,
+               <3 RK_PA3 RK_FUNC_GPIO &pcfg_pull_down>,
+               <3 RK_PA4 RK_FUNC_GPIO &pcfg_pull_down>,
+               <3 RK_PA5 RK_FUNC_GPIO &pcfg_pull_down>,
+               <3 RK_PA6 RK_FUNC_GPIO &pcfg_pull_down>,
+               <3 RK_PA7 RK_FUNC_GPIO &pcfg_pull_down>;
+       };
    };

@@ -140,21 +140,18 @@ &sdio {
 };

 &sdmmc {
    max-frequency = <200000000>;
    no-sdio;
    no-mmc;
    bus-width = <4>;
    cap-mmc-highspeed;
    cap-sd-highspeed;
    disable-wp;
-   pinctrl-names = "default";
+   pinctrl-names = "normal", "idle";
    pinctrl-0 = <&sdmmc0_clk &sdmmc0_cmd &sdmmc0_det &sdmmc0_bus4>;
```

```
+   pinctrl-1 = <&sdmmc0_idle_gpios &sdmmc0_det>;
    vmmc-supply = <&vcc3v3_sd>;
    vqmmc-supply = <&vccio_sd>;
    status = "okay";
 };
```